



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

Week 4 - Day 6 Report

Internship Program — Batch 2026

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1. Introduction

Experiment No. 18 focused on Model Deployment and Monitoring, addressing the critical realization that deploying a machine learning model to production is only the beginning of its lifecycle. Once a model is live and serving predictions, it must be continuously monitored to ensure it continues to perform well over time. Models can degrade silently as the data they process changes, a phenomenon known as data drift. Model monitoring involves tracking model performance, detecting anomalies, and identifying when retraining is needed. This experiment explored the complete deployment pipeline and the monitoring practices essential for maintaining production ML systems. Tools including MLflow for experiment tracking, Evidently AI for drift monitoring, and GitHub Actions for CI/CD were examined.

2. Objectives

- Understand the complete model deployment pipeline from training to production.
 - Learn how experiment tracking tools like MLflow capture and organize model metadata.
 - Understand the concepts of data drift and model monitoring.
 - Learn how drift detection helps identify when models need retraining.
 - Explore CI/CD practices for automating testing and deployment.
 - Understand the tools used in production ML environments.
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3. Task Description

Experiment No. 18 involved studying the end-to-end deployment flow and the continuous monitoring required to keep models performing well in production. The MLflow framework was explored for experiment tracking, understanding how training runs are logged with parameters, metrics, and model artifacts.

The concept of data drift was examined, understanding how changes in the input data distribution can degrade model performance over time. Monitoring techniques with tools like Evidently AI were studied to understand how drift is detected and alerted.

CI/CD practices were also explored, understanding how automated testing and deployment enable fast, safe releases. The role of GitHub Actions in automating workflows was examined, including how changes are tested and deployed automatically on code pushes.

4. Work Done

Experiment No. 18 began with understanding the full deployment flow from a trained model to a live application accessed by end users. The importance of each stage was examined, including model validation, API development, containerization, and production deployment.

The MLflow experiment tracking framework was explored in detail, learning how to log parameters, metrics, and model artifacts for each training run. The ability to compare runs and identify the best performing models was demonstrated through examples.

Data drift and concept drift were studied as key challenges in production ML systems. Understanding how changes in data distributions can silently degrade model accuracy was essential for appreciating the importance of continuous monitoring.

The Evidently AI monitoring tool was examined, understanding how it detects drift by comparing reference datasets with live production data. Alerting mechanisms and retraining triggers were studied to understand how monitoring systems enable proactive maintenance.

GitHub Actions for CI/CD was explored, understanding how automated workflows can run tests on every code push, ensuring code quality and preventing deployment of broken systems.

5. Challenges Faced

One of the main challenges in Experiment No. 18 was understanding the complexity of production ML systems and the many components that must work together for reliable operation. The deployment pipeline, monitoring infrastructure, and CI/CD automation all require careful design and coordination.

Another challenge involved understanding how data drift manifests in real-world systems and how it can be distinguished from normal data variation. Learning about statistical methods for drift detection required careful study of data distributions and monitoring metrics.

6. Key Learnings

Experiment No. 18 provided valuable knowledge of model deployment and production monitoring practices. I learned how MLflow enables reproducible experiments by tracking parameters, metrics, and models for every training run, making it easy to identify the best performing models and reproduce results.

The concept of data drift was particularly insightful, understanding how models can degrade over time as the data they process changes. This highlighted the importance of continuous monitoring and the need for automated retraining pipelines.

I also learned about CI/CD practices and how they enable fast, reliable releases in ML systems. The automation of testing and deployment through tools like GitHub Actions reduces manual errors and accelerates the development cycle.

Understanding the complete production ML stack, from experiment tracking to deployment to monitoring, provided a comprehensive view of what it takes to build reliable ML systems at scale.

7. Conclusion

Experiment No. 18 successfully demonstrated the essential practices for keeping ML systems reliable and performing well in production. By understanding the full lifecycle from experiment tracking through deployment and monitoring, I gained insight into the operational realities of production AI systems.

The experiment highlighted that building a model is only a small part of the overall ML lifecycle. The infrastructure for tracking, deploying, monitoring, and maintaining models requires significant attention and careful design. The skills and concepts learned provide a strong foundation for working on production ML systems where reliability, reproducibility, and continuous improvement are paramount.

Intern Signature:



Supervisor:

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Name:

Date: 25 July 2026

Designation: